Self-assessment Worksheet (Worksheet 4)

Do I understand this weeks material?

Real Analysis Topics	Tick the box when you feel confident with each topic
• Understand the definition of a series, convergent series and divergent series.	☐
• Know and utilise the different series tests to identify whether a given series is convergent or divergent.	☐
• Understand and know the definition of absolute convergence and conditional convergence.	☐

Exercises and Questions

Unless stated otherwise, you may assume any results proven in lecture.

Exercises and questions are organised and classified into three groups:

- **Starter Questions** - these questions will help cement the basic techniques.
- **Challenge Questions** - these questions are more challenging, designed to test your deeper understanding of the topic.
- **Extension Questions** - questions to try yourself in your own time to extend your knowledge and understanding.

Starter Qs

Have a go at these "warm-up" questions to get you started.

Question 1 I can do this question ☐

- Prove that the following series are divergent

(a) $\sum_{k=1}^{\infty}(-1)^{k+1}$.

(b) $\sum_{k=1}^{\infty}\frac{2k+1}{k+3}$.

(c) $\sum_{k=1}^{\infty}\frac{1}{2}$.

Question 2 I can do this question ☐

- Discuss the convergence of the following series

(a) $\sum_{k=1}^{\infty}\frac{\cos(k)}{k^2+1}$.

(b) $\sum_{k=1}^{\infty}\frac{k^2\sin(k)}{k^4+3k^2+1}$.

(c) $\sum_{k=1}^{\infty}\frac{(-1)^k k^2}{2^k}$.

Question 3 I can do this question ☐

- Using the ratio test, discuss the convergence of the following series:

(a) $\sum_{k=1}^{\infty}\frac{6^k}{k!}$.

(b) $\sum_{k=1}^{\infty} \frac{k3^k}{4^k}$.

(c) $\sum_{k=1}^{\infty} \frac{5^k}{4^k k}$.

(d) $\sum_{k=1}^{\infty} \frac{1}{k}$.

Question 4 I can do this question ☐

- Using the comparison and P-series test, discuss the convergence of the following series:

(a) $\sum_{k=1}^{\infty} \frac{k^2}{k^3+1}$.

(b) $\sum_{k=1}^{\infty} \frac{k}{k^3+1}$.

(c) $\sum_{k=1}^{\infty} \frac{(k+1)\sin^2(k)}{k^3+3k+1}$.

Challenge Qs

Now you are feeling more confident try these more challenging questions.

Question 5 I can do this question ☐

- Comment on the convergence/divergence of the following series:

(a) $\sum_{k=1}^{\infty} \frac{(-1)^k}{\sqrt{k}}$.

(b) $\sum_{k=1}^{\infty} \frac{1}{k!}$.

(c) $\sum_{k=1}^{\infty} \frac{(-1)^k}{k^{1/3}}$.

(d) $\sum_{k=1}^{\infty} \frac{(-1)^k k}{k+7}$.

(e) $\sum_{k=1}^{\infty} (\sqrt{k+1} - \sqrt{k})$.

(f) $\sum_{k=53}^{\infty} \frac{\ln(k)}{\ln(\ln(k))}$.

(g) $\sum_{k=1}^{\infty} \frac{2}{k} + \frac{3}{k^2}$.

(h) $\sum_{k=1}^{\infty} \frac{\sin(k\pi/2)}{k}$.

Question 6 I can do this question ☐

- Prove Theorem 3.5 from Chapter 3 of the notes.

Question 7 I can do this question ☐

- When proving the alternating series test, it was claimed that for $a_k \geq 0$ for each $k \in \mathbb{N}$ it followed that

$$|a_{m+1} - a_{m+2} + a_{m+3} - a_{m+4} + \cdots \pm a_n| \leq |a_{m+1}|$$

for all $n > m$. Prove this claim.

Question 8 I can do this question ☐

- Prove that if the series $\sum_{k=1}^{\infty} a_k$ diverges, then the series $\sum_{k=1}^{\infty}(ca_k)$ diverges, with $c \in \mathbb{R} \setminus \{0\}$.

Question 9 I can do this question ☐

- Consider a series $\sum_{k=1}^{\infty} a_k$ with $a_k \geq 0$ for each $k \in \mathbb{N}$ and suppose $L = \lim((a_k)^{1/k})$.
  (a) Suppose $L < 1$. Defining $\varepsilon = \frac{1-L}{2}$ and $L_1 = L + \varepsilon$, prove that there exists an $N \in \mathbb{N}$ such that $(a_k)^{1/k} < L_1$ for all $k > N$.
  (b) Using part (a), prove that $\sum_{k=1}^{\infty} a_k$ converges.
  (c) Now suppose $L > 1$. Defining $\varepsilon = \dfrac{L-1}{2}$, and $L_2 = L - \varepsilon$, prove that there exists an $N \in \mathbb{N}$ such that $(a_k)^{1/k} > L_2$ for all $k > N$.
  (d) Using part (c), prove that $\sum_{k=1}^{\infty} a_k$ diverges.

Question 10 I can do this question ☐

- (a) Prove that if (ka_k) converges to a non-zero real number, then the series $\sum_{k=1}^{\infty} a_k$ diverges. Give a counterexample to prove the converse is false.
  (b) Prove that if $(k^2 a_k)$ converges, then the series $\sum_{k=1}^{\infty} a_k$ converges. Give a counterexample to prove the converse is false.

Question 11 I can do this question ☐

- Give an example of a divergent series $\sum_{k=1}^{\infty} a_k$ and a convergent series $\sum_{k=1}^{\infty} b_k$ such that $a_k \leq b_k$ for all $k \in \mathbb{N}$.

Question 12 I can do this question ☐

- Discuss the convergence/divergence/absolute convergence and conditional convergence of the following series:
  (a) $\displaystyle\sum_{k=1}^{\infty}(-1)^k \frac{\sqrt{k+1} - \sqrt{k}}{k^p}$ with $p > 0$.
  (b) $\displaystyle\sum_{k=1}^{\infty} \frac{1}{p^k - q^k}$ with $0 < q < p$.
  (c) $\displaystyle\sum_{k=1}^{\infty} \frac{1}{p^k}$ with $p \in \mathbb{R}$.

Question 13 I can do this question ☐

- Let (a_k) and (b_k) be two sequences such that $a_k = b_{k+1} - b_k$ for each $k \in \mathbb{N}$. Prove that $\sum_{k=1}^{\infty} a_k$ converges if and only if $\lim(b_k)$ exists and additionally show that

$$\sum_{k=1}^{\infty} a_k = \lim(b_k) - b_1.$$

Question 14 I can do this question ☐

- Prove that if $a_k > 0$ for all $k \in \mathbb{N}$ and $\sum_{k=1}^{\infty} a_k^2$ converges, then $\sum_{k=1}^{\infty} a_k/k$ converges.

Question 15 I can do this question ☐

- (a) Let (a_k) be a monotone decreasing sequence with $\lim(a_k) = 0$. Prove that $a_k \geq 0$ for each $k \in \mathbb{N}$.
  (b) Hence, prove that if (a_k) is a monotone decreasing sequence and $\sum_{k=1}^{\infty} a_k$ is convergent, then $\lim(ka_k) = 0$.

Extension Qs

These questions are optional extension problems so you can gain more practice and confidence with these topics.

Question 16 I can do this question ☐

- Consider the series $\sum_{k=1}^{\infty} a_k$. Then the series $\sum_{k=1}^{\infty} b_k$ is called a **rearrangement** of $\sum_{k=1}^{\infty} a_k$ if there exists a bijective function $\sigma : \mathbb{N} \to \mathbb{N}$ such that $b_{\sigma(k)} = a_k$.

  (a) Prove that if a series converges absolutely, then any rearrangement of this series converges to the same limit.

  (b) Consider the series $\sum_{k=1}^{\infty} a_k$ with $a_k \geq 0$ for each $k \in \mathbb{N}$ and $\sum_{k=1}^{\infty} a_k$ diverges to $+\infty$. Prove that any rearrangement of this series diverges to $+\infty$.

Question 17 I can do this question ☐

- Let $p : \mathbb{N} \to \mathbb{N}$ be a bijective function such that

$$p(k) < p(l), \text{ if } k < l.$$

Consider the series $\sum_{k=1}^{\infty} a_k$. Then the series $\sum_{k=1}^{\infty} b_k$ is said to be **obtained from** $\sum_{k=1}^{\infty} a_k$ **by inserting parentheses** if

$$b_1 = a_1 + a_2 + \cdots + a_{p(1)}$$

and

$$b_{k+1} = a_{p(k)+1} + a_{p(k)+2} + \cdots + a_{p(k+1)},$$

for $k \in \mathbb{N}$. Equivalently, it could also be said that $\sum_{k=1}^{\infty} a_k$ is **obtained from** $\sum_{k=1}^{\infty} b_k$ **by removing parentheses** .

  (a) Prove that if a series $\sum_{k=1}^{\infty} a_k$ is convergent and another series $\sum_{k=1}^{\infty} b_k$ is obtained from $\sum_{k=1}^{\infty} a_k$ by inserting parentheses, then the series $\sum_{k=1}^{\infty} b_k$ is convergent.

  (b) Prove or disprove the converse of part (a).

End Material

Review and Summary

Make sure you have fully understood and read the notes. Once the solutions for this worksheet have been released, check your answers and make a note of areas of concern. If you are still unsure then email your lecturer or drop by in office hours and we will be happy to help!

Completing Worksheets: Tips

- Proceed carefully and systematically from beginning to end making sure to note theorems and properties utilised.
- Work together with other students, discuss ideas and concepts.
- Prioritise carefulness over speed.
- Include enough information in your answers that you can easily understand your thinking when you come back to review the worksheets at a later date.
- Do ask for help if you get stuck!